

Semantic Pulse: A Neuro-Symbolic Middleware for Selective State Hydration on the Orbital Edge

Daniel Kleebinder
Distributed Systems Group
TU Wien
Vienna, Austria
d.kleebinder@dsg.tuwien.ac.at

Index Terms—orbital edge computing, vector-symbolic architecture, time-varying graphs, serverless workflows, scheduling

I. MOTIVATION

Recent years have shown a rapid increase of satellite launches to Low Earth Orbit (LEO), with numbers going beyond 16,000 in early 2026, which is an increase of around 20% since 2025 [2]. This trend is going to increase even more drastically in the upcoming years with planned LEO constellations such as *Amazon Leo* [17] with 3,000 additional satellites and Chinese governmental filings at the International Telecommunication Union (ITU) with over 200,000 satellites [27].

The rise of mega-constellations leads to a new kind of far-edge computing – Orbital Edge Computing (OEC) – which has the potential of becoming a globally available low-latency source of computation [8], [19], [25], extending classical “bent-pipe” satellite architectures. However, the extreme conditions in space and the high physical velocities, at which satellites operate, incur challenges that, in combination, are fundamentally different from what we encounter down on earth.

A. Downlink Bottleneck & Latency

A downlink bottleneck occurs, when data generated on the satellite that needs to be processed down on earth exceeds its downlink capabilities. The downlink capabilities are not only determined by the bandwidth but also by the time a satellite is visible to a certain ground station. This might also lead to locality imposed latency where data has to reside on the satellite until in range of an appropriate ground station. Increasing the number of ground stations is not always an option, since maintenance and licensing the necessary frequencies imposes significant cost [1], [13], [23], [24].

B. High Packet Loss and Network Churn

The highly dynamic nature of satellite constellations inevitably leads to network churn due to a constantly changing topology of Inter-satellite Links (ISLs). Furthermore, the impact of radiation, solar activities and “space weather” in general has been studied extensively since the beginning of the space age showing not only potential packet loss, but even satellite outages [3]–[5], [18], [29]. A hostile environment like

space requires dedicated software and network hardening to create a resilient computing continuum [7], [16].

C. Stationarity

When talking about edge computation in LEO constellations, the issue of stationarity inevitably arises. Recent work has shown use cases, where the locality of the data source is strictly bound to the computability of a problem. This includes peer-to-peer networks and natural disaster analysis [6], [21], [22]. Inherently, stationarity can be seen from two different perspectives: (1) from an earth bound view, where clients need a stable uplink and satellites need to appear stationarity and (2) from a space bound view, where computation on satellites can only be performed, when data sources are in proximity. From a systems point of view, both of these perspectives are inherently difficult to realize due to the highly dynamic nature of satellites.

II. SKETCHING THE IDEA

This work seeks to address the challenges previously described by proposing the idea of an *Ambient Holographic State*, which aims to create an ambient (i.e., static and location-aware) global data field, that incorporates holographic data representations for increased fault tolerance, resiliency and robustness against packet loss.

Figure 1 outlines the idea of ambient holographic state by introducing a wildfire natural disaster use case. Either edge devices on the ground that are equipped with appropriate sensors or sensor satellites detect natural disasters such as wildfires and encode location together with a label (e.g., *wildfire detected*) into an event. This event is bundled into the local state representation and delta updates are transmitted using high-speed ISLs and merged into each satellite’s local state, ensuring eventual consistency. Because information in a gossip-based network spreads gradually, satellites located closer to data sources receive notifications about changes and new data more quickly.

This requires the underlying data structure to be highly resilient to space weather induced bit-flips (a wildfire must be a wildfire, no matter if a single bit has been flipped or not), resource-efficient, enabling execution on highly constrained platforms (such as satellites), and commutative, in order to handle events that arrive out of order.

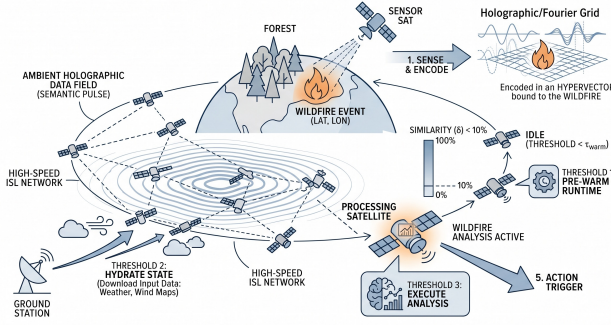


Fig. 1. Sensors on ground and sensor satellites detect a potential wildfire. They encode the label L_{fire} and the location X and Y into an event H_e that is propagated through the network and integrated into the local state of each satellite $\mathcal{H}$. As satellites get closer to the origin of the event H_e (i.e., δ increases), they might pre-warm (fetch heavy state from ground stations or other satellites, etc.) serverless workflows that registered for wildfire events, allowing for an appropriate heads-up start.

Notice, that the signal strength of an underlying event (i.e., the satellite quickly gets closer to the wildfire) can be used to *pre-warm* serverless workflows on a satellite. This includes loading heavy state from a near-by ground station before heading over the wildfire, where no ground stations are available or loading state through other satellites close by. Concluding the outline of the overall idea, this leads us to the following potential research questions:

- RQ1 How must an ambient holographic middleware be architected to enable a decentralized ambient data field that balances spatial resolution with information persistence in the 3D continuum?
- RQ2 How do orbital topology dynamics and gossip frequency influence the stochastic convergence of holographic state signals across a highly dynamic time-varying topology?
- RQ3 To what extent can an ambient holographic representation maintain state consistency under high degrees of packet loss and node churn?
- RQ4 How can the threshold $\tau_{prewarm}$ be dynamically tuned to balance the trade-off between hydration success rate and energy waste in power-constrained LEO nodes?

III. ANALYTICAL FORMAL MODEL

This section briefly introduces some formal models that are used to outline the idea of an ambient holographic state using Vector Symbolic Architecture (VSA) applied to LEO satellite constellations.

A. Serverless workflows

Research has shown the potential for serverless functions for orbital computing workflows [22], [26]. Serverless workflows consist of functions that depend on each other, where one functions output state is the input of the next downstream function. Thus, a directed acyclic graph based formal model

is used where $W = (F, E)$ with F being the set of serverless functions and $E = F \times F$ the set of connections between functions. This is the basic analytical model for (serverless) workflows that can be used to model Service-level Objectives (SLOs).

B. The periodic time-varying nature of satellite constellations

The periodic nature of LEO satellite network topologies can be formalized by using Time-Varying Graphs (TVGs) [11]. Let $\mathcal{G} = (N, E, L, \mathcal{T}, \rho, \zeta)$ be a TVG where N is the set of computing nodes (i.e., LEO satellites), E the set of edges defined as $E \subseteq N \times N \times L$ and L being an alphabet that defines labels containing properties such as ISL-availability or bandwidth. Then $\rho : E \times \mathcal{T} \rightarrow \{0, 1\}$ defines the presence function of a given edge and $\zeta : E \times \mathcal{T} \rightarrow \mathbb{T}$ being its *latency* function that varies in time. Following these definitions, the periodicity of the edges in TVG is defined as:

$$\forall e \in E, \forall t \in \mathcal{T}, \forall k \in \mathbb{N} : \rho(e, t) = \rho(e, t + kp) \quad (1)$$

Where $p \in \mathbb{T}$ defines the interval. Similar to paths in a graph, a *Journey* $\mathcal{J} = \{(e_1, t_1), \dots, (e_k, t_k)\}$ defines a time-varying path from one node to another. Using this definition, the latency for state transfer between two nodes in the network is defined as follows:

$$\zeta(\mathcal{J}) = \sum_{(e, t) \in \mathcal{J}} \zeta(e, t) \quad (2)$$

Introducing the definition of the latency SLO S_{ij} for the combination of serverless functions (f_i, f_j) :

$$\zeta(\mathcal{J}) \leq S_{i,j} \quad \forall \mathcal{J} \in \mathcal{J}_{\mathcal{G}}^*, \forall (f_i, f_j) \in W \quad (3)$$

Where $\mathcal{J}_{\mathcal{G}}^*$ defines the set of all journeys in a TVG. This notation can be used to encompass different kinds of SLOs while preserving the underlying time-varying nature of satellite constellations.

C. Holographic State using Hypervectors

Hyperdimensional Computing (HDC) emerged as a field of study from neuroscience, which assumes that the brains perceptive representation and computation are all performed in a hyperdimensional vector space $\mathbb{V} \in \mathbb{R}^d$ (where $d \geq 1000$) [15]. Assume two random bipolar hypervectors R which represents the color *red* and T which represents the shape *triangle*. Given these two representations, we can define the following operations: (i) A *binding* operation, denoted by $V = R \otimes T$, which can be interpreted as a composite that represents a *red triangle*. (ii) An *unbinding* operation, denoted by $R = V \oslash T$, which can be interpreted as to query the color of the triangle and (iii) a *bundling* operator $R \oplus T$ which represents a collection of similar concepts (e.g., *red and triangle typically go together*) [10].

HDC is inherently resilient due to its property of *smearing* information across a hypervector. If one bit is flipped, the state does not become invalid. Instead, it can be reconstructed from corrupted or incomplete information, thus being *holographic*.

1) *Topology-aware Binding*: Using this semantic, an *Ambient Holographic Event* $H_e \in \mathbb{V}$ is defined as the binding of semantic, spatial, and temporal components:

$$H_e = V_{semantic} \otimes V_{space} \otimes V_{time} \quad (4)$$

Where $\otimes$ denotes the element-wise multiplication (binding) operator. To create the *Ambient Holographic State* $\mathcal{H}_t$ at time t , we define the superposition of all distributed events via the bundling operator $\oplus$:

$$\mathcal{H}_t = (\alpha \mathcal{H}_{t-1}) \oplus \sum_{e \in \Delta \mathcal{E}_t} H_e \quad (5)$$

Here, $\alpha \in (0, 1]$ represents the *temporal decay factor* (which might mess with the commutative property of the ambient state), and $\Delta \mathcal{E}_t$ is the set of new events detected by the constellation in the current orbital time-step.

When a satellite updates either its local representation of the global ambient holographic state or its location information, it queries the state using the cosine similarity function δ :

$$\delta(\mathcal{H}_t, Q) = \frac{\mathcal{H}_t \cdot Q}{\|\mathcal{H}_t\| \|Q\|} \quad (6)$$

Where Q defines the spatial- and time-aware query on the state $\mathcal{H}_t$ as follows:

$$Q = V_{semantic} \otimes V_{space}(\text{sat}) \otimes V_{time}(\text{now}) \quad (7)$$

Notice that on common hardware, using the *Hamming distance* instead of the cosine similarity function might yield much better performance. The Hamming distance is defined as follows [9], [14]:

$$\Delta(\mathcal{H}_t, Q) = |\{j \in \{1, \dots, D\} \mid \mathcal{H}_{t,j} \neq Q_j\}| \quad (8)$$

The similarity function δ always returns a stochastic probability for an encoded value. This enables the satellite to pre-warm serverless workflows when the probability exceeds a given $\tau_{prewarm}$ and to trigger the workflow for some given τ :

$$\delta(\mathcal{H}_t, Q) \geq \tau_{prewarm} \quad (9)$$

2) *Fractional Power Encoding*: Since the VSA requires a discrete binary representation for efficient transmission and operation using binary operators, we cannot encode decimal values such as latitude (48.2082°) or longitude (16.3738°) in the state hypervector. Leveraging Fractional Power Encoding (FPE), a continuous frequency domain base vector can be defined, where decimal number encoding enables querying the state hypervector with higher stochastic probability the closer the satellite gets to an $\mathcal{H}_e$. The gradient with which the stochastic probability of $\delta(\mathcal{H}_t, Q)$ declines, is defined by the kernel $K(x, y)$ – to be defined in detail. This representation is called the Fourier Holographic Reduced Representation (FHRR) [10], [12], [28].

To map a continuous location $x \in \mathbb{R}$ into a hyperdimensional space, we first define a random base vector in the complex frequency domain. Let D be the dimensionality of the hypervector. We define a phase vector $\Phi = [\phi_0, \phi_1, \dots, \phi_{D-1}]$, where each phase is sampled uniformly from a circle:

$$\phi_k \sim \mathcal{U}(-\pi, \pi) \quad (10)$$

To ensure the final spatial vector strictly consists of real numbers (a requirement for standard VSA thresholding), we must enforce Hermitian conjugate symmetry on these phases. For $k = 1, \dots, \frac{D}{2} - 1$:

$$\phi_{D-k} = -\phi_k \quad (11)$$

This gives us our fixed base vector $\mathbb{B}$ in the complex frequency domain:

$$\mathbb{B} = [e^{i\phi_0}, e^{i\phi_1}, \dots, e^{i\phi_{D-1}}] \quad (12)$$

To encode the specific location x , we raise the base vector to the fractional power of x . In the complex exponential form, this mathematically equates to multiplying the phases by x according to the Fourier shift theorem. The encoded frequency-domain vector $\mathbb{V}_f(x)$ is:

$$\mathbb{V}_f(x) = \mathbb{B}^x = [e^{i(\phi_0 \cdot x)}, e^{i(\phi_1 \cdot x)}, \dots, e^{i(\phi_{D-1} \cdot x)}] \quad (13)$$

Because ϕ_k represents an angle, multiplying it by x rotates the phase. A continuous change in x results in a continuous rotation around the unit circle, inherently granting the system infinite theoretical resolution and perfect circular topology.

We cannot compute VSA bundling operations efficiently in the complex frequency domain. We use the Inverse Discrete Fourier Transformation (IDFT) to translate the phase-rotated vector back into the spatial (computational) domain. The continuous spatial hypervector $\mathcal{V}_c(x) \in \mathbb{R}^D$ has elements v_n defined as:

$$v_n(x) = \frac{1}{D} \sum_{k=0}^{D-1} e^{i(\phi_k \cdot x)} \cdot e^{i \frac{2\pi}{D} kn} \quad (14)$$

Because we enforced Hermitian symmetry in equation 11, all imaginary components perfectly cancel out during this summation, guaranteeing that $\mathcal{V}_c(x)$ is a strictly real-valued vector containing the interference pattern of the shifted frequencies. Parseval's theorem guarantees isometry and ensures similarity between vectors across transformations.

For efficient transmission over optical ISLs and extremely fast $O(1)$ XOR-based similarity comparisons, the continuous vector is projected into a dense bipolar space $\{-1, 1\}^D$. We apply the signum function element-wise:

$$\mathcal{H}_{space}(x) = \text{sgn}(\mathcal{V}_c(x)) \quad (15)$$

Where the $\text{sgn}(v)$ function is defined as follows:

$$\text{sgn}(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ -1 & \text{if } v < 0 \end{cases} \quad (16)$$

Resulting in a discrete bipolar space hypervector $\mathcal{H}_{space}(x)$ that can be encoded as part of an event or as part of a query Q . Alternatively, the piecewise $\text{sgn}(v)$ function could also directly map to a discrete binary space.

3) *Coordinate Interference Problem*: On earth and other spherical celestial bodies, locations are typically encoded using two distinct coordinates, longitude $[-180^\circ, 180^\circ]$ stretching from east to west (denoted as x) and latitude $[-90^\circ, 90^\circ]$ stretching from north to south (denoted as y). The most straightforward approach to creating a hypervector V_{space} that encodes both x and y is using the binding operator $\otimes$:

$$V_{space} = \mathcal{V}_c(x) \otimes \mathcal{V}_c(y) \quad (17)$$

Although binding is an extremely fast operation for binary hypervectors, this inevitably leads to coordinate bleeding because (i) the $\otimes$ operator is commutative and (ii) the complex frequency base vector $\mathbb{B}$ is the same for both coordinates. This means that a phase shift of n (i.e., $e^{i(\phi \cdot n)}$) leads to the same result for longitude and latitude. Therefore, we have to sample two random complex frequency base vectors $\mathbb{B}^x$ and $\mathbb{B}^y$ for longitude and latitude. Due to hyper-dimensionality of these two base vectors, they will inherently be almost perfectly orthogonal to each other. The encoding for both longitude and latitude is the complex element-wise multiplication of the phases:

$$V_{space} = \mathbb{B}^x \odot \mathbb{B}^y \quad (18)$$

Notice, that in the complex frequency domain, multiplying two phasors is the same as adding their phases:

$$e^{i\phi} \cdot e^{i\psi} = e^{i(\phi+\psi)} \quad (19)$$

This leads to the following definition for the space hypervector in the complex frequency domain:

$$\mathbb{V}_f(x, y) = \left[e^{i(x\phi_0 + y\psi_0)}, \dots, e^{i(x\phi_{D-1} + y\psi_{D-1})} \right] \quad (20)$$

The rest of the transformation back to the continuous signal domain remains the same using the IDFT.

4) *Wraparound Coordinates*: Longitude wraps around when exceeding its bounds (i.e., $180^\circ = -180^\circ$). This is not the case for latitude, since latitude always increases from south to north pole. We can use the properties of the complex frequency domain and the sampling from the unit circle to encode this particular property by adjusting the sampling of random values for $\mathbb{B}^x$:

$$\phi_k = \frac{2\pi \cdot m_k}{360^\circ} \quad (21)$$

Where m_k is a random integer between $[1, M_{max}]$. The value of M_{max} determines the kernel size. Higher values lead

to quicker gradient drop-off rates and vice versa because a phase $e^{i(x\phi)}$ being more susceptible to larger values of x . Determining the appropriate M_{max} is an optimization problem depending on the event that has to be registered and the pre-warm duration of the serverless workflow.

IV. POTENTIAL PITFALLS

This section briefly discusses potential pitfalls and tradeoffs FPE and VSA impose in general.

A. Communication Overhead

HDC trades resiliency for payload size. It inherently relies on hyperdimensional vectors, specifically $\mathbb{V} \in \mathbb{R}^d$ where $d \geq 1000$ this imposes a communication overhead when transmitting the state vector. A thorough evaluation must be performed that compares the ambient holographic state approach against traditional payload routing mechanisms. Failover strategies and packet loss must be considered.

B. Implications of the Decay Factor (α)

The decay factor α is of importance for the computation of the global time-constraint state $\mathcal{H}_t$. Otherwise, older state $\mathcal{H}_{t-1}$ will not lose its impact on the new state. This means, that once a wildfire has been detected, it will inevitably create noise throughout the entire lifespan of $\mathcal{H}_t$ reducing precision of the similarity function $\delta(\mathcal{H}_t, Q)$. The implications of different decay factors must be thoroughly studied.

C. Serverless Workflow Execution

The execution of serverless workflows (and serverless functions) that are executed (in parts) on a satellite constellation must be studied extensively. The impact of the ambient holographic state and τ that is used to pre-warm a serverless function on a satellite to prepare it for execution when the satellite is above a certain geographic location (or event) must be compared to existing scheduling strategies such as *Databelt* [20].

REFERENCES

- [1] European Space Agency-funded projects reach new performance level in groundwork for optical LEO to GEO data relays – ESA CSC: Connectivity & Secure Communications (Aug 2024), <https://connectivity.esa.int/archives/news/european-space-agency-funded-projects-reached-new-performance-level-groundwork-optical-leo-geo-data-relays>
- [2] Space Environment Statistics · Space Debris User Portal (Jan 2026), <https://sdup.esoc.esa.int/discosweb/statistics/>
- [3] Allen, J., Sauer, H., Frank, L., Reiff, P.: Effects of the March 1989 solar activity. *Eos, Transactions American Geophysical Union* **70**(46), 1479–1488 (1989). <https://doi.org/10.1029/89EO00409>, <https://onlinelibrary.wiley.com/doi/abs/10.1029/89EO00409>, _eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/89EO00409>
- [4] Astafyeva, E., Yasyukevich, Y., Maksikov, A., Zhivetiev, I.: Geomagnetic storms, super-storms, and their impacts on GPS-based navigation systems. *Space Weather* **12**(7), 508–525 (2014). <https://doi.org/10.1002/2014SW001072>, <https://onlinelibrary.wiley.com/doi/abs/10.1002/2014SW001072>, _eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1002/2014SW001072>

- [5] Baker, D.N.: Satellite Anomalies due to Space Storms. In: Daglis, I.A. (ed.) *Space Storms and Space Weather Hazards*, pp. 285–311. Springer Netherlands, Dordrecht (2001). https://doi.org/10.1007/978-94-010-0983-6_11
- [6] Bhattacharjee, D., Kassing, S., Licciardello, M., Singla, A.: In-orbit Computing: An Outlandish thought Experiment? In: *Proceedings of the 19th ACM Workshop on Hot Topics in Networks*. pp. 197–204. HotNets '20, Association for Computing Machinery, New York, NY, USA (Nov 2020). <https://doi.org/10.1145/3422604.3425937>, <https://dl.acm.org/doi/10.1145/3422604.3425937>
- [7] Bhattacharjee, D., Singla, A.: Network topology design at 27,000 km/hour. In: *Proceedings of the 15th International Conference on Emerging Networking Experiments And Technologies*. pp. 341–354. ACM, Orlando Florida (Dec 2019). <https://doi.org/10.1145/3359989.3365407>, <https://dl.acm.org/doi/10.1145/3359989.3365407>
- [8] Bhosale, V., Bhardwaj, K., Gavrilovska, A.: Toward Loosely Coupled Orchestration for the LEO Satellite Edge (2020)
- [9] Bookstein, A., Kulyukin, V., Raita, T.: Generalized Hamming Distance. *Information Retrieval* **5** (Oct 2002). <https://doi.org/10.1023/A:1020499411651>
- [10] Bremer, A., Orchard, J.: Improved Cleanup and Decoding of Fractional Power Encodings (Nov 2024). <https://doi.org/10.48550/arXiv.2412.00488>
- [11] Casteigts, A., Flocchini, P., Quattrociocchi, W., Santoro, N.: Time-Varying Graphs and Dynamic Networks (Feb 2012). <https://doi.org/10.48550/arXiv.1012.0009>, <http://arxiv.org/abs/1012.0009>, arXiv:1012.0009 [cs]
- [12] Frady, E.P., Kleyko, D., Kymn, C.J., Olshausen, B.A., Sommer, F.T.: Computing on Functions Using Randomized Vector Representations (in brief). In: *Proceedings of the 2022 Annual Neuro-Inspired Computational Elements Conference*. pp. 115–122. NICE '22, Association for Computing Machinery, New York, NY, USA (2022). <https://doi.org/10.1145/3517343.3522597>, <https://dl.acm.org/doi/10.1145/3517343.3522597>
- [13] Furutanpey, A., Zhang, Q., Raith, P., Pfandzelter, T., Wang, S., Dustdar, S.: FOOL: Addressing the Downlink Bottleneck in Satellite Computing With Neural Feature Compression. *IEEE Transactions on Mobile Computing* **24**(8), 6747–6764 (Aug 2025). <https://doi.org/10.1109/TMC.2025.3544516>, <https://ieeexplore.ieee.org/document/10897922/>
- [14] Hamming, R.W.: *Coding and information theory* (2nd ed.). Prentice-Hall, Inc., USA (Feb 1986)
- [15] Kanerva, P.: Hyperdimensional Computing: An Introduction to Computing in Distributed Representation with High-Dimensional Random Vectors. *Cognitive Computation* **1**(2), 139–159 (Jun 2009). <https://doi.org/10.1007/s12559-009-9009-8>, <http://link.springer.com/10.1007/s12559-009-9009-8>
- [16] Kassing, S., Bhattacharjee, D., Águas, A.B., Saethre, J.E., Singla, A.: Exploring the "Internet from space" with Hypatia. In: *Proceedings of the ACM Internet Measurement Conference*. pp. 214–229. IMC '20, Association for Computing Machinery, New York, NY, USA (2020). <https://doi.org/10.1145/3419394.3423635>, <https://dl.acm.org/doi/10.1145/3419394.3423635>
- [17] Kohnstamm, T.: Everything you need to know about Amazon Leo, Amazon's satellite broadband network (Nov 2025), <https://www.aboutamazon.com/news/innovation-at-amazon/what-is-amazon-project-kuiper>
- [18] Lakhina, G.S., Tsurutani, B.T.: Geomagnetic storms: historical perspective to modern view. *Geoscience Letters* **3**(1), 5 (Feb 2016). <https://doi.org/10.1186/s40562-016-0037-4>, <https://doi.org/10.1186/s40562-016-0037-4>
- [19] Ma, X., Xu, M., Li, Q., Li, Y., Zhou, A., Wang, S.: Visions of Edge Computing in 6G. In: Ma, X., Xu, M., Li, Q., Li, Y., Zhou, A., Wang, S. (eds.) *5G Edge Computing: Technologies, Applications and Future Visions*, pp. 179–202. Springer Nature, Singapore (2024). https://doi.org/10.1007/978-981-97-0213-8_9
- [20] Marcelino, C., Guelmino, L., Pusztai, T., Nastic, S.: Databelt: A Continuous Data Path for Serverless Workflows in the 3D Compute Continuum (Aug 2025). <https://doi.org/10.48550/arXiv.2508.15351>, <http://arxiv.org/abs/2508.15351>, arXiv:2508.15351 [cs]
- [21] Pfandzelter, T., Bermbach, D.: Celestial: Virtual Software System Testbeds for the LEO Edge. In: *Proceedings of the 23rd ACM/IFIP International Middleware Conference*. pp. 69–81. Middleware '22, Association for Computing Machinery, New York, NY, USA (2022). <https://doi.org/10.1145/3528535.3531517>, <https://dl.acm.org/doi/10.1145/3528535.3531517>
- [22] Pfandzelter, T., Hasenburger, J., Bermbach, D.: Towards a Computing Platform for the LEO Edge. In: *Proceedings of the 4th International Workshop on Edge Systems, Analytics and Networking*, pp. 43–48. Association for Computing Machinery, New York, NY, United States (Apr 2021). <https://doi.org/10.1145/3434770.3459736>, <https://dl.acm.org/doi/epdf/10.1145/3434770.3459736>
- [23] Teodoro, A.C., Duarte, L.B.C.B.: The role of satellite remote sensing in natural disaster management. In: *Nanotechnology-Based Smart Remote Sensing Networks for Disaster Prevention*, pp. 189–216. Elsevier (Jan 2022). <https://doi.org/10.1016/B978-0-323-91166-5.00015-X>, <https://www.sciencedirect.com/science/article/pii/B978032391166500015X>
- [24] Vasisht, D., Shenoy, J., Chandra, R.: L2D2: low latency distributed downlink for LEO satellites. In: *Proceedings of the 2021 ACM SIGCOMM 2021 Conference*. pp. 151–164. SIGCOMM '21, Association for Computing Machinery, New York, NY, USA (Aug 2021). <https://doi.org/10.1145/3452296.3472932>, <https://dl.acm.org/doi/10.1145/3452296.3472932>
- [25] Wang, S., Li, Q.: Satellite Computing: Vision and Challenges. *IEEE Internet of Things Journal* **10**(24), 22514–22529 (Dec 2023). <https://doi.org/10.1109/IIOT.2023.3303346>, <https://ieeexplore.ieee.org/document/10213423/>
- [26] Wang, S., Li, Q., Xu, M., Ma, X., Zhou, A., Sun, Q.: Tiansuan Constellation: An Open Research Platform. In: *2021 IEEE International Conference on Edge Computing (EDGE)*. pp. 94–101 (Sep 2021). <https://doi.org/10.1109/EDGE53862.2021.00022>, <https://ieeexplore.ieee.org/document/9712016/>, iSSN: 2767-9918
- [27] Weissberger, A.: China ITU filing to put ~200K satellites in low earth orbit while FCC authorizes 7.5K additional Starlink LEO satellites – IEEE ComSoc Technology Blog (Jan 2026), <https://techblog.comsoc.org/2026/01/13/china-itu-filing-to-put-200k-satellites-in-orbit-fcc-authorizes-7-5k-additional-starlink-leo-satellites/>
- [28] Yeung, C., Zou, Z., Imani, M.: Generalized Holographic Reduced Representations (May 2024). <https://doi.org/10.48550/arXiv.2405.09689>, <http://arxiv.org/abs/2405.09689>, arXiv:2405.09689 [cs]
- [29] Zhang, S., He, L., Wu, L.: Statistical Study of Loss of GPS Signals Caused by Severe and Great Geomagnetic Storms. *Journal of Geophysical Research: Space Physics* **125**(9), e2019JA027749 (2020). <https://doi.org/10.1029/2019JA027749>, <https://onlinelibrary.wiley.com/doi/abs/10.1029/2019JA027749>, eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2019JA027749>